

AKASH JAPE

akashjape@gmail.com • (972)-806-2102 • <https://tinyurl.com/akashjdesignportfolio>

EDUCATION

Bachelor of Science in Mechanical Engineering

December 2027

The University of Texas at Austin, Materials Science Concentration, GPA: 3.86/4.00

Relevant courses: Machine Elements, Dynamic Systems & Controls, Mechatronics, Materials Processing, Mech. Behavior of Materials, Heat Transfer, Fluid Mechanics

EXPERIENCE

SharkNinja – Needham, MA

May 2026 – August 2026

Product Design Engineering Intern, Advanced Development

- Designed a user-operated retention mechanism for rotating and translating shaft; developed concept variants and validated through user demos and cross-functional testing with Product Integrity and Development teams
- Designed and prototyped a mechanical safety interlock for a 3-part assembly; poka-yoked assembly to a single correct orientation; validated for resilience to impact, water, and heat cycles
- Led new product ideation for novel countertop appliance; conducted food science research, culinary testing, and market studies assessing desirability, viability, and feasibility for presentation to cross-functional teams

Qorvo – Richardson, TX

May 2025 – August 2025

Process Engineering Intern

- Led DoE-driven investigation of epoxy dispensing defects in automated lid-attach assembly; isolated needle geometry as root cause and validated improvements via optical metrology, reducing dispense error by 96%
- Designed automated engineered fluid dispensing (EFD) process replacing manual epoxy application for scalable production; validated process equivalence using cross-sectioning, optical measurement, and JEDEC MSL testing
- Applied 5S methodology to design and additively manufacture ESD-compliant thermoplastic tooling; validated improvements through user testing and time-motion studies, reducing excess movement and cycle time

Texas Inventionworks – Austin, TX

January 2024 – Present

Vice President • Prior Roles: Operations Executive Lead, Student Technician

- Led 60+ staff across operations, training, and technical services supporting 3,000+ students; scaled access to manufacturing and rapid prototyping resources including machining, additive manufacturing, and laser cutting
- Developed and taught mechanical and electrical hardware workshops guiding students through full product design cycles including CAD modeling, prototype fabrication, design validation testing, and design iteration
- Managed scheduling for 60+ employees; led community outreach and initiatives with external stakeholders

Nanoscale Design and Manufacturing Laboratory – Austin, TX

June 2024 – January 2025

Undergraduate Research Assistant – Micro-scale Selective Laser Sintering

- Designed machine components; developed python-LabVIEW integration

PROJECTS & ACTIVITIES

National Medical Device Make-A-Thon – San Antonio, TX

February 2024

Second Place – Women's Health Challenge

- Identified redesign opportunity for gynecological tool through systematic research on patient discomfort; analyzed prior art and existing improvement attempts to inform design requirements
- Developed patient-administered speculum concept with medical-grade silicone coating, addressing ergonomic and psychological pain points; developed SolidWorks CAD model and FDM-printed functional prototype

SKILLS

Mechanical Design & Fabrication: Concept ideation, human-centered design, GD&T, tolerance analysis, FDM/SLA/SLS, machining, laser cutting, EFD dispensing, carbon fiber assembly, soldering, rapid prototyping, DFM/DFA

Process & Manufacturing: DoE, Lean Principles, 5S methodology, A3 problem solving, root cause analysis, failure analysis

Software: SolidWorks, Python, LabVIEW, MATLAB, Java (Oracle SE 8 Certified), Jira, Microsoft Suite

Languages: English, Marathi (fluent); French (conversational)